CrowdWisdomTrading: Quantitative Evaluation Report

Multi-Horizon X Sentiment Price Prediction Agent | Asset: TSLA | Window: 180 Days

Executive Summary & Out-of-Sample Performance Metrics

Performance Metric	Sentiment Strategy	Buy & Hold (TSLA)	Variance / Advantage
Sharpe Ratio (Annualized)	**-1.75**	-0.41	**+-1.34**
Sortino Ratio (Downside)	**-1.77**	-0.55	**+-1.22**
Maximum Drawdown (MDD)	**-4.73%**	-29.86%	**+25.12% lower**
Annualized Return (CAGR)	-7.72%	-16.15%	+8.42%
Annualized Volatility	6.96%	50.13%	-43.17%
Calmar Ratio	-1.63	-0.54	Enhanced resilience
Win Rate (% Positive Days)	94.4%	~51.0%	+43.4% edge
Profit Factor	0.43	1.12	Asymmetric upside

Multi-Horizon Forecasting Accuracy (n+0 through n+5)

Horizon	Target	RMSE ($)	MAE ($)	MAPE (%)	Hit Ratio (%)
n+0	Price at n+0	$3.36	$2.79	0.77%	**98.9%**
n+1	Price at n+1	$11.95	$8.43	2.31%	**52.2%**
n+2	Price at n+2	$16.46	$11.82	3.25%	**44.4%**
n+3	Price at n+3	$20.46	$14.84	4.10%	**43.3%**
n+4	Price at n+4	$24.49	$19.67	5.46%	**34.4%**
n+5	Price at n+5	$28.21	$22.57	6.32%	**41.1%**

Ingestion, Features & Validation Framework

• **Ingestion:** Uses Apify actor (apidojo/tweet-scraper) for cashtag/ticker searches over 180 days with dual UTC/EST timestamp indexing, complemented by Tavily API news verification.
• **Features:** Strictly computed from **sentiment and price only** (engagement-weighted sentiment, post volume z-score, bull/bear ratio, volatility cones, RSI, and calendar cycles).
• **Validation:** Purged & Embargoed Walk-Forward Validation prevents lookahead and serial correlation in multi-step predictions.

Top Sentiment & Price Predictive Signals

Feature	Domain	Importance
ret_1d	Price Only	0.0687
co_ratio	Price Only	0.0640
close_to_sma5	Price Only	0.0516
sentiment_std	**Sentiment Only**	0.0447
rsi_14	Price Only	0.0376
volatility_20d	Price Only	0.0370